

Homework — 1.4.1 Data Types

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.4.1 lesson. *SHOW ALL WORKING.*

Part A — This week's topic (~14 marks)

1. Convert the denary number **156** to 8-bit unsigned binary. **[2]** (AO2)

Working space:

2. Convert the binary number **11001010** to hexadecimal. **[2]** (AO2)

Working space:

3. Convert the hexadecimal number **6F** to denary. **[2]** (AO2)

Working space:

4. Using 8-bit **two's complement**, calculate **52 – 37** by adding the two's complement of 37. Show the binary working and state clearly what happens to the final carry. **[4]** (AO2)

Working space:

5. The 8-bit unsigned binary number 01001100 is shifted using an **arithmetic right shift of 2 places** (treat the value as signed two's complement). Give the resulting bit pattern and state the arithmetic effect. [2] (AO2)

Working space:

6. A games developer says, "Real numbers and integers are basically the same, so it doesn't matter which data type I pick." Explain **one** reason a programmer must choose the correct data type, referring to how each is represented. [2] (AO1)

Working space:

Part B — Retrieval: keep it fresh (~6 marks)

7. State the role of the **MAR** (Memory Address Register) in the fetch–decode–execute cycle. (1.1) [1] (AO1)

Working space:

8. Describe how **virtual memory** allows a computer to run a program that is larger than the installed RAM. (1.2) [3] (AO1)

Working space:

9. State **one** difference between **lossless** and **lossy** compression, and give a suitable use for each. (1.3) [2] (AO1)

Working space:
